MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The number of two-digit numbers divisible by 3 is: [1]

(A) 25 (B) 30 (C) 32 (D) 33

Q2. If the nth term of an AP is $(2n+1)$, then the sum of its first n terms is: [1]

(A) $n(n-2)$ (B) $n(n+2)$ (C) $n(n+1)$ (D) n^2

Q3. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ is: [1]

(A) p (B) $-p$ (C) -1 (D) 1

Q4. If $a_n = 3 - 4n$, then the sequence is an AP with common difference: [1]

(A) 3 (B) 4 (C) -4 (D) -3

Q5. **Assertion (A):** The 10th term from the end of the AP $4, 9, 14, \ldots, 254$ is 209.
Reason (R): The nth term from the end of a finite AP with last term l is $l-(n-1)d$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

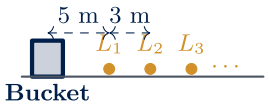
Q6. Find the common difference of an AP whose first term is 5 and the sum of its first four terms is half the sum of the next four terms. [2]

Q7. In an AP, if $a_n = 9 - 5n$ is the nth term, find the sum of its first 15 terms. [3]

Q8. Find the 4th term of the AP whose sum of first n terms is $S_n = 2n^2 + 3n$. [3]

Q9. Find the sum of first 20 terms of an AP in which $d = 7$ and the 20th term is 143. [3]

Q10. CASE STUDY: Annual Sports Day Lemon-and-Bucket Race



In an annual school sports meet, a lemon-and-bucket race is organized. A collection bucket is fixed at the starting line. Ten fresh lemons are placed in a straight line extending from the bucket.
The first lemon is placed 5 m from the bucket, and every subsequent lemon is placed 3 m further apart from the preceding one. Each competitor starts from the bucket, runs to pick up the nearest lemon, runs back to deposit it into the bucket, and repeats this routine until all 10 lemons are safely gathered.

Based on the above mathematical scenario, answer the following questions:

(i) Find the total round-trip distance run by a competitor to collect the first lemon and return it to the bucket. [1]

(ii) Find the round-trip distance run to collect the second lemon. [1]

(iii) Write down the AP representing the round-trip distances and calculate the total cumulative distance run by a competitor to gather all 10 lemons. [2]

— OR —

What distance does the competitor run to collect only the 10th (final) lemon? [2]

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